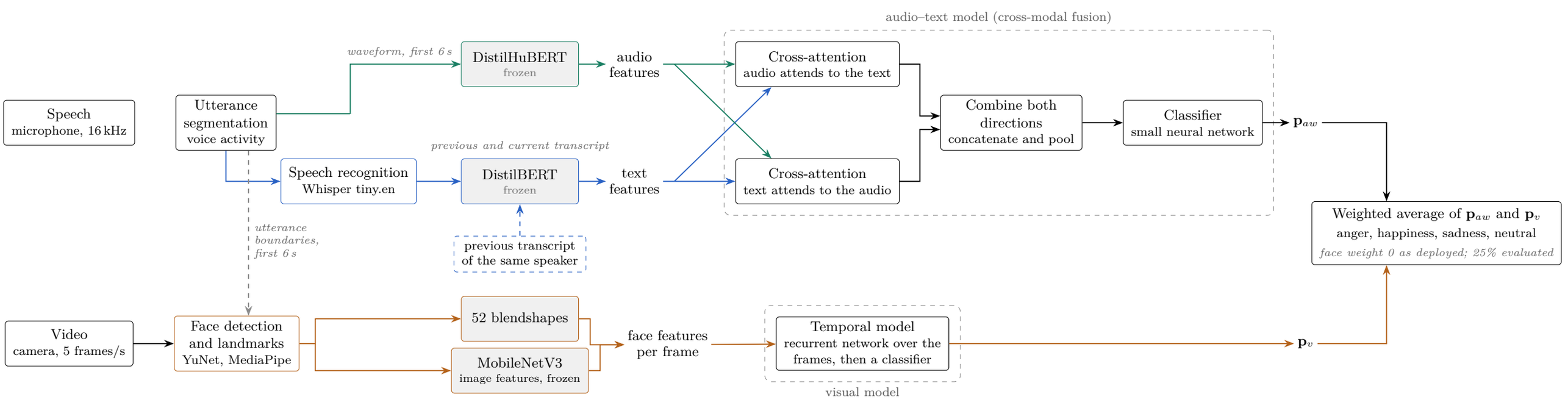

audio–text model (cross-modal fusion)
Speech
microphone, 16 kHz
Utterance
segmentation
voice activity
waveform, first 6 s
DistilHuBERT
frozen
audio
features
Cross-attention
audio attends to the text
Cross-attention
text attends to the audio
Combine both
directions
concatenate and pool
Classifier
small neural network
$\mathbf{p}_{aw}$
previous and current transcript
Speech recognition
Whisper tiny.en
DistilBERT
frozen
text
features
previous transcript
of the same speaker
utterance
boundaries,
first 6 s
Weighted average of $\mathbf{p}_{aw}$ and $\mathbf{p}_v$
anger, happiness, sadness, neutral
face weight 0 as deployed; 25% evaluated
Video
camera, 5 frames/s
Face detection
and landmarks
YuNet, MediaPipe
52 blendshapes
MobileNetV3
image features, frozen
face features
per frame
Temporal model
recurrent network over the
frames, then a classifier
$\mathbf{p}_v$
visual model